



FORWARD DEPLOYED ENGINEER

Location: Bristol | Type: Fulltime/Contract

About Newpage Solutions

Newpage Solutions is a global digital health innovation company helping people live longer, healthier lives. We partner with life sciences organisations which include, pharmaceutical, biotech and healthcare leaders, to build transformative AI and data driven technologies addressing real-world health challenges.

From strategy and research to UX design and agile development, we deliver and validate impactful solutions using lean, human-centered practices.

We are proud to be a **'Great Place to Work®' certified** company for the last three consecutive years. We also hold a top **Glassdoor rating** and are named among the "Top 50 Most Promising Healthcare Solution Providers" by CIOReview. As an organisation, we foster creativity, continuous learning and inclusivity, creating an environment where bold ideas thrive and make a measurable difference in people's lives.

Your Mission

We are hiring Forward Deployed Engineers who treat AI as the substrate of how software gets built—not a tool to be cautious of, not something they are "exploring," but the medium they work in.

You will sit close to clients and product leaders, reframe vague problems into something concrete, and ship working software end-to-end. You won't wait for a refined backlog, a PM in the middle, or a separate platform team. You will shape the idea, build the thing, and stand it up in production yourself.

This is a builder-first individual-contributor role for engineers who live at the current edge of AI development and work fluently with Claude Code, Cursor, agents, eval harnesses, MCP, and modern TypeScript and Python.





What You'll Do

Problem / Opportunity Discovery

- Sit with a business or clinical leader and reframe an idea or problem into something concrete and buildable.
- Know what to build by the end of the conversation; have a working prototype to react to by the end of the week.
- Partner closely with product, design, and client stakeholders to translate ambiguous ideas into software that ships.
- Demo live without a slide deck. Reframe problems out loud. Don't get stuck waiting for someone else to make the decision.
- Lead POCs, innovation sprints, and internal research experiments to validate emerging AI techniques.

Build (fast) with AI

- When the brief is clear, head down and produce.
- Build modular backends in Python or TypeScript aligned with clean architecture, OOP, SOLID, and domain-driven design.
- Create fullstack applications, APIs, agents, workflows, and similar systems using frameworks such as Next.js, React, FastAPI, Fastify, FastMCP, and Hono.
- Architect and ship production-grade agentic applications using LangGraph, AutoGen, Claude Agent SDK, OpenAI Assistants, or your own orchestration layer.
- Integrate frontier and self-hosted LLMs (Claude, GPT, Gemini, open-weight models) with tools, data, and external systems through MCP and custom connectors.
- Apply RAG techniques where they actually help: vector databases (Pinecone, Chroma, Weaviate, pgvector), hybrid retrieval with Elasticsearch or Solr, and BM25 + similarity search.
- Work across relational, document, key-value, and graph stores as the problem demands; use event-driven patterns where they fit, not by default.
- Design prompt and context engineering frameworks that optimize accuracy, repeatability, cost, and latency.
- Use AI-assisted development tools (Claude Code, GitHub Copilot, Cursor, Codex) through structured workflows, native instructions, templates, and sub-agents—with discipline and review.
- Fine-tune or adapt models where the problem genuinely calls for it.



Test, Deploy, Productionize

- Spin up the infra, write the evals, wire up the MCP servers, deploy the agents, and harden the bits that survive contact with real users.
- Deploy on AWS, Azure, Cloudflare, or Vercel using containerization (Docker, Kubernetes) or serverless—chosen for fit, not preference.
- Treat evals as a first-class discipline: hands-on harnesses, not theoretical frameworks. Build with a clear-eyed view of where current AI tooling helps and where it falls short.
- Apply engineering practices that hold up in production: TDD, secrets management and rotation, SAST/DAST, structured logging, metrics, tracing, and automated CI/CD (GitHub Actions, Jenkins).
- Own what you build end-to-end, including the infrastructure and operations that keep it running.
- Mentor others on system design, agentic patterns, and AI engineering best practices.

What You Bring

- 3+ years relevant experience building production applications using AI / agentic development approaches—fullstack applications, agents, workflows, MCPs, and more.
- Hands-on experience with agents, not just prompted models. You have wired tools to a model and let it run multi-step using LangGraph, AutoGen, Claude Agent SDK, OpenAI Assistants, or your own orchestration.
- Active, structured use of AI-assisted development tools (Claude Code, Cursor, GitHub Copilot) with demonstrable workflows, sub-agents, skills, and innovative approaches.
- Strong Python or TypeScript, with OOP, SOLID, 12-factor application development, and microservice architecture. You've built Next.js applications, FastAPI services, and similar.
- End-to-end implementation experience with vector databases, retrieval pipelines, and eval harnesses.
- Cloud-native deployment experience across at least one of AWS, Azure, Cloudflare, or Vercel—with Docker, Kubernetes, and GitHub Actions.
- A no-compromise attitude on clean code, TDD, security, observability, scalability, performance, and cost.



- A deep working understanding of how LLMs behave—and where they break—and how to optimize accuracy, latency, and cost.
- Clear writing and a willingness to reframe problems in conversation rather than wait for someone else to define them.
- A real, recent trail of built things: GitHub, a portfolio, side projects, indie tools, or OSS contributions.
- A founder's mindset and genuine appetite for ambiguous, high-impact technical challenges.
- Bachelor's or Master's in Computer Science, Machine Learning, or a related technical discipline.

Bonus Skills / Experience

- Public writing, talks, or threads about building with AI.
- MLOps and model serving experience (BentoML, MLflow, Vertex AI, SageMaker).
- Streaming and batch ingestion pipelines (Spark, Airflow, Beam, Glue).
- Healthcare or life sciences domain exposure.
- AWS Professional certification or other relevant industry certifications.

What We Offer

At Newpage, we're building a company that works smart and grows with agility, where driven individuals come together to do work that matters. We offer:

- A people-first culture - Supportive peers, open communication and a strong sense of belonging
- Smart, purposeful collaboration - Work with talented colleagues to create technologies that solve meaningful business challenges
- Balance that lasts - We respect your time and support a healthy integration of work and life
- Room to grow - Opportunities for learning, leadership and career development, shaped around you
- Meaningful rewards - Competitive compensation that recognises both contribution and potential





Ready to Apply?

Let's build the future of health together. Apply below or reach out to:
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